

Course Code	21CSS201T	Course Name	COMPUTER ORGANIZATION AND ARCHITECTURE	Course Category	S	ENGINEERING SCIENCES	L	T	P	C
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Pre-requisite Courses	Nil	Co- requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	School of Computing	Data Book / Codes / Standards	Nil		Nil

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	understand the Fundamentals of computers, Memory operations and Addressing Modes
CLR-2:	know about Functions of Arithmetic and Logic unit
CLR-3:	explore the Operations of Control Unit, Execution of Instruction and Pipelining
CLR-4:	classify the Need for Parallelism, Multicore and Multiprocessor Systems
CLR-5:	understand the Concepts and functions of Memory unit, I/O unit

Course Outcomes (CO):	At the end of this course, learners will be able to:
CO-1:	identify the computer hardware and how software interacts with computer hardware
CO-2:	apply Boolean algebra as related to designing computer logic, through simple combinational and sequential logic circuits
CO-3:	examine the detailed operation of Basic Processing units and the performance of Pipelining
CO-4:	analyze concepts of parallelism and multi-core processors
CO-5:	classify the memory technologies, input-output systems and evaluate the performance of memory system

Unit-1 – Introduction to Number System	Introduction to Number System and Logic Gates: Number Systems- Binary, Decimal, Octal, Hexadecimal: Codes- Grey, BCD, Excess-3, ASCII, Parity; Binary Arithmetic- Addition, Subtraction, Multiplication, Division using Sign Magnitude, 1's complement, 2's complement, BCD Arithmetic; Logic Gates-AND, OR, NOT, NAND, NOR, EX-OR, EX-NOR.	12 Hour
Unit-2 - Architectures	Basic structure of computers: Functional Units of a computer, Operational concepts, Bus structures, Memory addresses and operations, assembly language, Instructions, Instruction sequencing, Addressing modes. Case study: 8086.	12 Hour
Unit-3 - Design of ALU	De Morgan's Theorem, Adders, Multiplier – Unsigned, Signed, Fast, Carry Save Addition of summands; Division–Restoring and Non-Restoring; IEEE 754 Floating point numbers and operations	12 Hour
Unit-4 - Control Unit	Basic processing unit, ALU operations, Instruction execution, Branch instruction, Multiple bus organization, Hardwired control, Generation of control signals, Micro-programmed control; Pipelining: Basic concepts of pipelining, Performance, Hazards-Data, Instruction and Control, Influence on instruction sets.	12 Hour
Unit-5 - Parallelism	Need, types, applications and challenges, Architecture of Parallel Systems-Flynn's classification; ARM Processor: The thumb instruction set, Processor and CPU cores, Instruction Encoding format, Memory load and Store instruction, Basics of I/O operations. Case study: ARM 5 and ARM 7 Architecture.	12Hour

Learning Resources	1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Computer Organization, 5th ed., McGraw-Hill, 2015 2. AiHwang, Faye A. Briggs, Computer Architecture and Parallel Processing, 3rd ed., McGraw-Hill, 2016 3. Ghosh T.K., Computer Organization and Architecture, 3rd ed., Tata McGraw-Hill, 2011 4. P. Hayes, Computer Architecture and Organization, 3rd ed., McGraw-Hill, 2015.	5. William Stallings, Computer Organization and Architecture—Designing for Performance, 10th ed., Pearson Education, 2015 6. David A. Patterson and John L. Hennessy, Computer Organization and Design—A Hardware/Software Interface, 5th ed., Morgan Kaufmann, 2014
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Learning Assessment		Continuous Learning Assessment (CLA)					
	Bloom's Level of Thinking	Formative		Life-Long Learning CLA-2 (10%)		Summative Final Examination (40% weightage)	
		CLA-1 Average of unit test (50%)		Theory	Practice	Theory	Practice
		Theory	Practice				
Level 1	Remember	30%	-	30%	-	30%	-
Level 2	Understand	30%	-	30%	-	30%	-
Level 3	Apply	20%	-	20%	-	20%	-
Level 4	Analyze	20%	-	20%	-	20%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
Total		100 %		100 %		100 %	

Course Designers	
Experts from Industry	Internal Experts
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